Microbes in Production of Fine Chemicals

Antibiotics
Drugs
Vitamins
Amino Acids

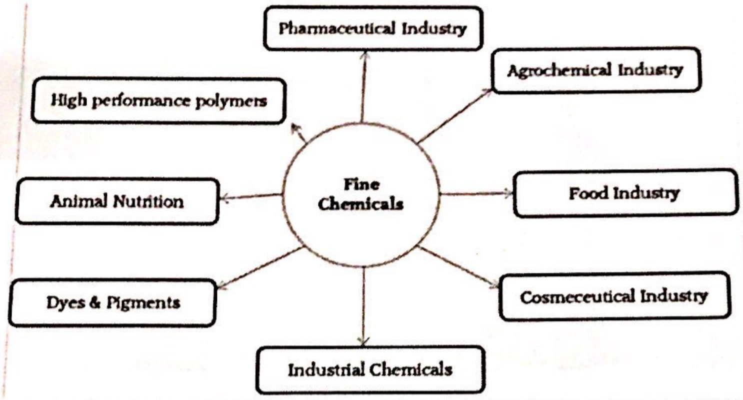
Fine Chemicals

- Fine chemicals are single pure substances produced in small to medium quantities and have a high value.

* to inject chemicals into human bodies it must be highly pure

Fine Chemicals

- Fine chemicals are single pure substances produced in small to medium quantities and have a high value.



Antibiotics

- **Antibiotics** are small sized chemicals produced by 2/3 antibiotics microorganisms and which in low concentrations are capable of inhibiting the growth of, or killing, other microorganisms

- Most antibiotics are secondary metabolites produced by filamentous fungi and bacteria, particularly the actinomycetes (in ~~secondar~~ stationary phase)

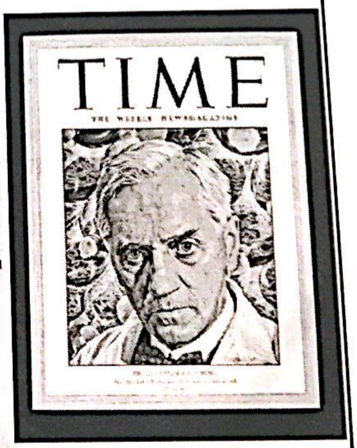
- **Antimicrobial agent:** Chemical that kills or inhibits the growth of microorganisms

Ehrlich's Magic Bullets

↗ first antibiotic produced

Fleming & Penicillin

FIG. 1.—Photograph of a culture-plate showing the dissolution of staphylococcal colonies in the neighbourhood of a penicillium colony.



* the discovery of penicillin was by chance, the scientist was originally studying staphylococcus, due to contamination there was a fungal growth that killed the staph around it

* fleming discovered the fungi that produced the penicillin but 2 other scientists purified it

Antibiotic Spectrum of Activity

TABLE 20.2 **The Spectrum of Activity of Antibiotics and Other Antimicrobial Drugs**

Prokaryotes				Eukaryotes			
Mycobacteria*	Gram-Negative Bacteria	Gram-Positive Bacteria	Chlamydias, Rickettsias†	Fungi	Protozoa	Helminths	Viruses
		← Penicillin →		← Ketoconazole →		← Niclosamide → (tapeworms)	
← Streptomycin	Streptomycin →				← Mefloquine → (malaria)		
							← Acyclovir →
						← Praziquantel → (flukes)	
	← Tetracycline	Tetracycline	Tetracycline →				
← Isoniazid →							

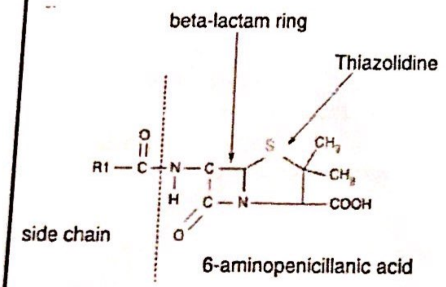
*Growth of these bacteria frequently occurs within macrophages or tissue structures.
†Obligately intracellular bacteria.

* antibiotic spectrum: what targets are effected by antibiotics
* some types of antibiotics can target both G+ and G- bacteria

* why are broad spectrum antibiotics ~~are~~ more important than narrow spectrum?
because in some cases the antibiotics need to be given before knowing the type of bacteria because some cases need to be treated fast for high chances of death

Antibiotics discovered from fungi

- *Penicillium* and *Acremonium* spp. predominantly synthesize β-lactam antibiotics

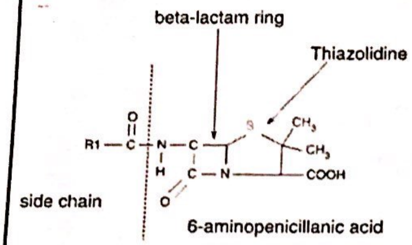


Penicillin produced by *Penicillium spp*

2/30/2

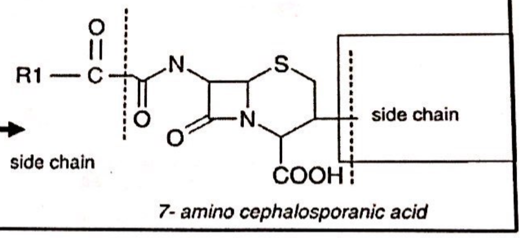
Antibiotics discovered from fungi

- *Penicillium* and *Acremonium* spp. predominantly synthesize β-lactam antibiotics

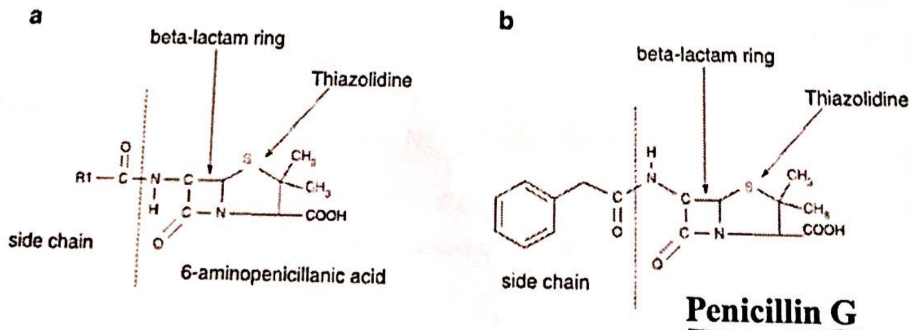


Penicillin produced by *Penicillium spp*

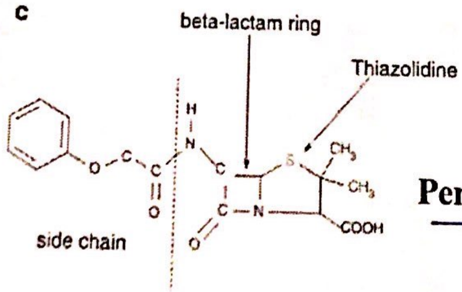
Cephalosporins produced by *Acremonium spp*



Penicillins



Penicillin G



Penicillin V

the difference is in the side chains

Production of Penicillins

Production medium → Production fermentation
Inoculum development → Production fermentation

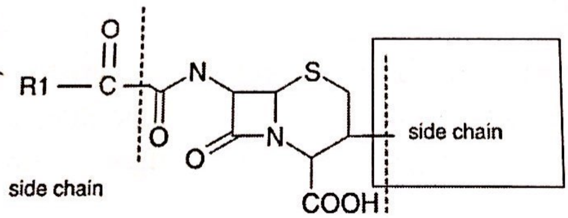
Production fermentation (a fed-batch process)
↓
Broth clarification
e.g. rotary vacuum filtration → Washed mycelium
↓
Solvent extraction of cell-free broth
e.g. amyl acetate
↓
Extract purification (supernatant)
e.g. activated charcoal treatment and filtration
↓
Crystallization of antibiotic
e.g. addition of sodium acetate
↓
Recovery of penicillin crystals
e.g. filtration
↓
Crystal washing
e.g. anhydrous ethanol wash and filtration
↓
Crystal drying
e.g. vacuum tray drier
↓ Natural penicillin
↓ Penicillin acylase treatment and the addition of a 'new' side chain → Semi-synthetic penicillin

*when you crystalize a protein it comes out pure

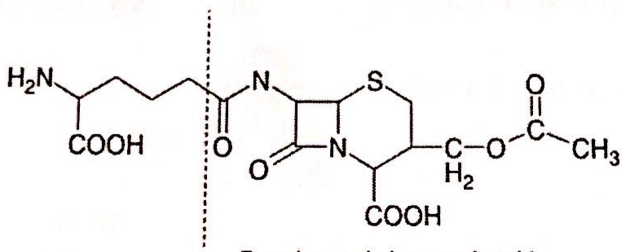
*when the bacteria becomes resistant to penicillin we make a semi-synthetic penicillin to fool it

Cephalosporins

20 different side chains ←



7- amino cephalosporanic acid



7- amino cephalosporanic acid

Cephalosporin C

β-lactams (Cell wall synthesis inhibitors)

1. **Natural penicillins (*Penicillium chrysogenum)***
   - Penicillin G and Penicillin V
   - Effective against Gram-positive bacteria (because they are cell wall synthesis inhibitors, they prevent the formation of peptidoglycan layer)
2. **Semi-synthetic penicillins (*Penicillium chrysogenum)***
   - Aminopenicillins — amino group added to penicillin
   - Penicillinase-resistant penicillins
   - Carboxypenicillins
   - Ureidopenicillins.

* G+ has a thick peptidoglycan layer while G- has a thin one

3. **Cephalosporins (*Cephalosporium acremonium)***
   - Broad spectrum (G+ and G−)
   *can be used in infections caused by several microorganisms
4. **Monobactams (*Chromobacterium violaceum*)**
   - Gram-negative bacteria

soil mycroorganisms

Actinomycetes in Antibiotic Discovery

- Of about 20,000 antibiotics produced via microorganisms, 45 % of antibiotics come from actinomycetes, 80 % of which come from a single genus Streptomyces
- Streptomycin was the first aminoglycoside class of antibiotic discovered by Selman A. Waksman and Albert Schatz
- *Streptomyces sp.* has been an important source of major classes of antibacterial drugs, namely tetracyclins, aminoglycosides, macrolides, chloramphenicol (acetamide), and β-lactams

*streptomyces also produce anticancer agents, immunosuppressors, antifungal agents, antiviral agents

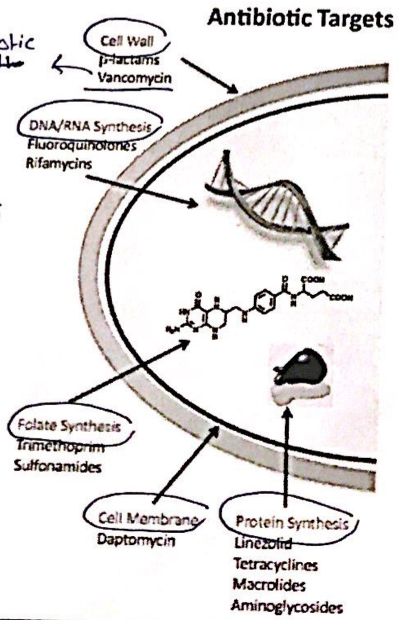
Microbial Sources of Antibiotics

Microorganism	Antibiotic	
Gram-Positive Rods		
Bacillus subtilis	Bacitracin	**Polypeptide antibiotic**
Bacillus polymyxa	Polymyxin	**Polypeptide antibiotic**
Actinomycetes		
Streptomyces nodosus	Amphotericin B	**Polyene (Antifungal)**
Streptomyces venezuelae	Chloramphenicol	**Chloramphenicols**
Streptomyces aureofaciens	Chlortetracycline and tetracycline	**Tetracyclines**
Streptomyces erythraeus	Erythromycin	**Macrolides**
Streptomyces fradiae	Neomycin	**Aminoglycides**
Streptomyces griseus	Streptomycin	**Aminoglycides**
Micromonospora purpureae	Gentamicin	**Aminoglycides**
Fungi		
Cephalosporium spp.	Cephalothin	**β-lactams**
Penicillium griseofulvum	Griseofulvin	
Penicillium notatum	Penicillin	**β-lactams**

Modes of Antimicrobial Action & Resistance

the last resort antibiotic
to treat MRSA

* MRSA : penecillin resistance staphylococcus aureas
(multi resistance)

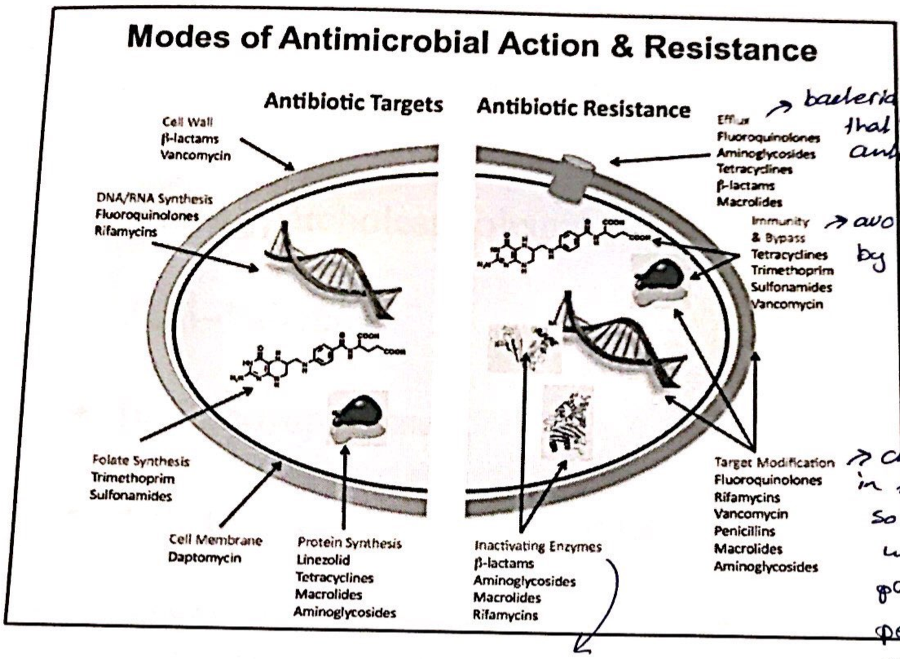


cell wall: prevent production of cell wall (G+)

DNA/RNA synthesis: prevent bacteria from replicating it's genetic materal so it dies

Folate synthesis: bacteria need to produce follic acid so it can replicate, some antibiotics inhibit this production

protein synthesis: if bacteria doesn't produce proteins it'll die

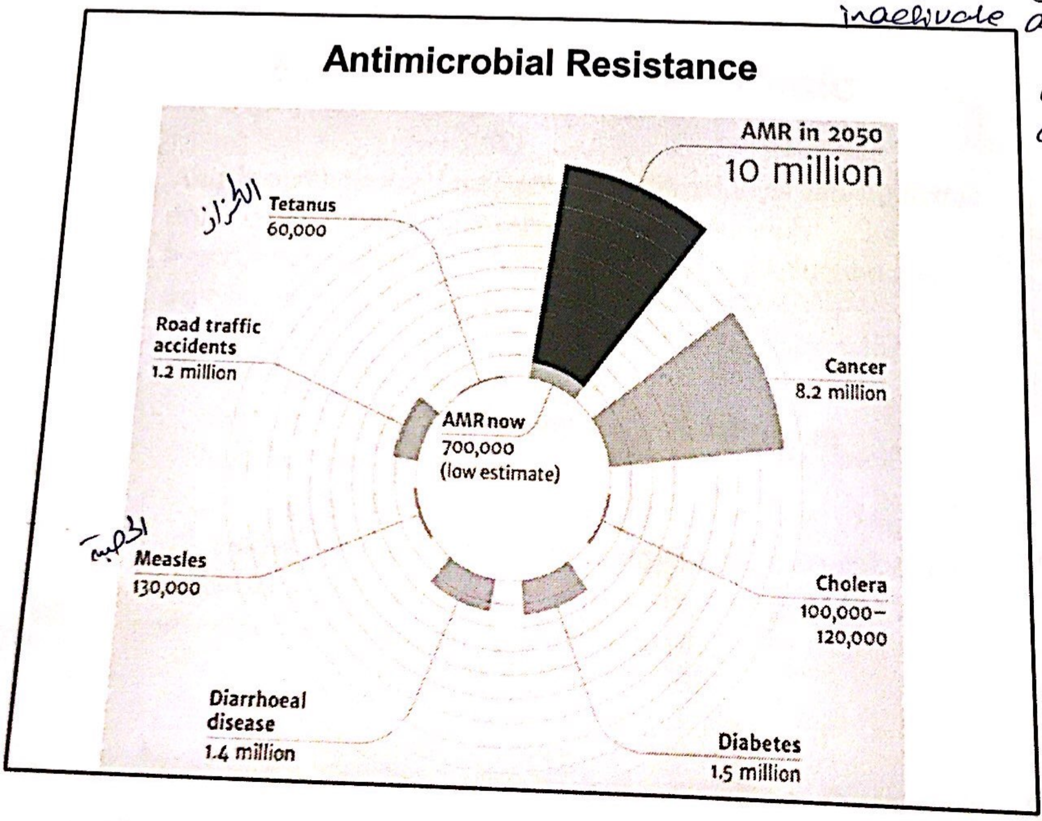


→ bacteria produce a pump that'll take out the antibiotic if it gets in

→ avoid being recognized by immune system

→ change certain things in the cell wall structure so that the antibiotic will not recognize it, produce low affinity penicillin binding proteins so that it wouldn't be able to bind to penicillin

the enzymes that help the effect of antibiotics will be inactivated, produce enzymes that inactivate antibiotics, an enzyme that cut the side chain of the antibiotic



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الحصبة

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Health-products produced by microbes

- Anti-hypercholesterolemic
- Anti-diabetic
- Immunosuppressant
- Anti-tumor/Anti-cancer agents

*all produced as secondary metabolites in stationary phase

habity
ribic)

Anti-hypercholesterolemic

*cholestrol is very important and very vital
* liver and extra hepatic tissue (extra tissues outside the liver) produce cholesterol at night
*cholesterol can be taken also from animal food
*cholesterol is dangerous in the presence of fat (can cause hypertension)
* smoking lead to Arteriosclerosis تصلب الشرايين which leads to hypertension الضغط

- Anti-hypercholesterolemic are also referred to as anti-lipidemic drugs or statins and are prescribed to patients with hypercholesterolemia, i.e. excess cholesterol production and deposition.
- Examples:
  - Mevastatin from *Penicillium* and lovastatin from *Aspergillus* (fungi)
- Both inhibit the enzyme hydroxy-methylglutaryl co-enzyme A (HMG-CoA) reductase, one of the enzymes responsible for conversion of mevalonate into cholesterol.

* most amount of cholestrol is found in brain

*cholestrol high levels may be caused from a genetic disorder called ~~hyper~~ familial hyper cholesterolemia which can't be treated by reducing the amount of cholesterol uptake in diet but must be treated by Anti-hypercholesterolemic drugs which prevent the liver from producing extra cholesterol

Anti-diabetic

- Acarbose and Valiolamine are isolated from actinomycetes
- Inhibit the α-glucosidase enzymes present in the intestine to block the breakdown and absorption of oligosaccharide and polysaccharides (decrease the amout of free glucose)

*oligosaccharide: 3-5 saccharides compounds
polysaccharides: multi
these produce alot of sugar

*If we eat food but we don't have insulin in our bodies then the food will not be metabolized thus glucose will not be metabolized and it will not enter the cells so we'll have high sugar content

Immunosuppressants

- Immunosuppressive agents are substances that inhibit or prevent the activity of the immune system.
- Used in the transplantation of organs or tissues to prevent rejection, in the treatment of autoimmune disorders and non-autoimmune inflammatory conditions → hyperactivation of the immune system
- Examples: → complicated chemical structure produced by mold
  - Cyclosporine from *Tolypocladium inflatum* interferes with the activity and growth of T cells.

the major part of immune system that's responsible for rejection

Anti-tumor/Anti-cancer agents

- Mainly produced by *Streptomyces sp.* produced at metaphase

Drugs for Alzheimer's Dementia

(β-amaylaze accumalation)

- *Streptomyces griseofuscus* produces **phytostigmine**, which improves memory function in the brain of healthy humans as well as in those with Alzheimer's dementia.

* Alzheimer's drugs don't treat Alzheimer's but reduce the symptoms

Endophytic Microbes as Sources of Putative Phytochemicals

from plants

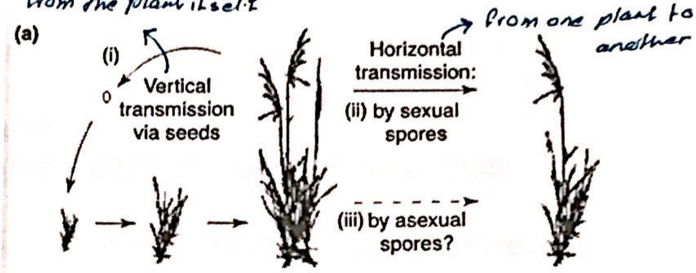
- Endophytes are bacteria, fungi, and actinomycetes, which spend part of their complete life cycle colonizing in healthy plant tissues inter- or intra-cellularly.
- Almost all vascular plants on earth harbor endophytic microbes
- Endophytes biosynthesize important plant compounds or phytochemicals

plant chemicals

Endophytic Microbes as Sources of Putative Phytochemicals

from the plant itself

(a)



(b)

	Plant	Fungus
Benefits	Increased: • Growth • Reproduction • Resistance	• Refuge • Nutrition • Transmission

TRENDS in Plant Science

Endophytic Microbes as Sources of Putative Phytochemicals

- *Taxomyces andreanae* produces the multi-billion dollar anti-cancer compound **Taxol** (generic name paclitaxel)
- Used to treat prostate, ovarian, breast, and lung cancer.
- Many approaches (e.g. plant cell culture technology or chemical synthesis for paclitaxel production) have been developed, but cost- effective bulk production is still not achievable, resulting in the high cost of drug

(it's still can't be produced in high amounts)

* drugs. cost : from idea → production ≈ 800,000,000 $

Microbial Synthesis of Vitamins

- Vitamins are micronutrients that are required by all organisms in trace quantities but that cannot be synthesized by mammals and are instead synthesized by microorganisms or plants.
- They are also used as food/feed additives and as therapeutic agents.
- Processed foods, feeds, pharmaceuticals, cosmetics, and chemicals contain extraneously added vitamins or vitamin- related compounds.

Microbial Synthesis of Vitamins

water soluble
C, B complexes
fat, ~~lipid~~ soluble
ADEK
(antioxidents)
(anti free-radicals)
(لا يمكن الاستغناء عنهم
indenspensable)

Vitamin E (Tocopherols)

- Photosynthetic microorganisms are known to accumulate detectable amounts of tocopherols
- The model system for genetic engineering for over-production of tocopherols is a cyanobacterium
  (we engineer cyanobacterium to produce lots of vitamin E)

Vitamin K (non protein factor)

- The major role of vitamin K is in blood coagulation
- Required for the prevention of bone loss and bone fractures in humans, antioxidant activity, and reducing the effects of Alzheimer's disease

Microbial Synthesis of Vitamins

β-Carotene (Provitamin A) (in carrots and egg yolk)

- Present in the chloroplast and chromoplasts of plants, photosynthetic bacteria, fungi, and microalgae.
- Microorganisms contribute to approximately 15 % of the total industrial production

* β-carotene is ~~pro~~ introduced in rice to produce it in poor contries (golden rice)

Vitamin B2

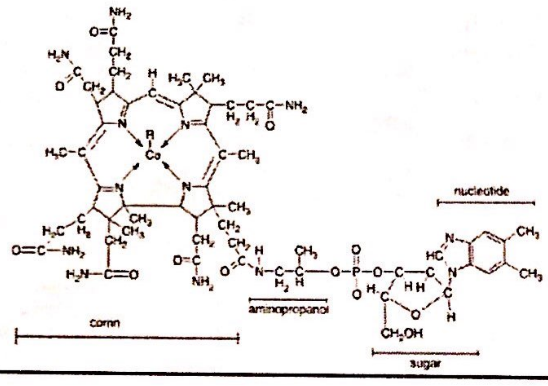
- Also known as riboflavin
- Plays an important role as a precursor to flavin mononucleotide (FMN) and flavin adenine dinucleotide (FAD), which function as coenzymes to a variety of enzyme catalyzed reactions in the intermediate metabolism (ex. TCA cycle / Krep's cycle)

Microbial Synthesis of Vitamins

Vitamin B12 B12 can't be synthesized by the body but must be obtained from outer sources

- Also known as cobalamin and performs a key role in the normal function of the brain and nervous system and in the formation of blood.
- Only bacteria and archaea have the enzymes required for its synthesis, although many foods are a natural source of B12 because of bacterial symbiosis

*B12 defeciency can cause bernecious anemia, these people have problems with B12 absorbtion



Microbial Synthesis of Vitamins

- Microorganisms utilized in biotransformation of therapeutic steroids that are used for the treatment of allergies, inflammation, skin diseases and as oral contraceptives

Table 11.3 Examples of some steroids whose production involves microbial biotransformation, and a selection of specific biotransformations

(a) Steroids			
Androgens	Testosterone		
Corticosteroids	Cortisone, hydrocortisone, prednisone and dexamethasone		
Oestrogens	Oestradiol and estrone		
Gestagens	Progesterone		
(b) Biotransformation	**Substrate**	**Product**	**Organism**
1-Dehydrogenation	Hydrocortisone	Prednisolone	Mainly fungi & mycobacteria
11 α-Hydroxylation	Progesterone	11 α-Hydroxyprogesterone	
11 β-Hydroxylation	Reichstein compound S	Hydrocortisone	
Side-chain cleavage	β-sitosterol	Androstadienedione	

* cortizon is produced by certain molds

Production of amino acids by microbes

Estimated annual production of important primary metabolites of microorganisms worldwide

Metabolite	Production (tons)	Market value (million dollars)
AMINO ACIDS		
L-Glutamate	1,000,000	3000
L-Lysine	800,000	915
L-Threonine	20,000	100
L-Aspartate	13,000	198
L-Isoleucine	400	43
NUCLEOTIDES		
5'-IMP + 5'-GMP	2500	350
ORGANIC ACIDS		
Citric acid	400,000	1400
VITAMINS		
B_{12}	3	100
C[a]	60,000	71
Riboflavin	2000	60

[a] Partially synthetic.
IMP, inosine monophosphate; GMP, guanosine monophosphate.
Based mainly on Demain, A. L. (2000). Small bugs, big business: the economic power of the microbe. *Biotechnology Advances*, 18, 499–514.

for lip health ←

L stand for levo

vitamin C partially synthetic ←

* amino acids can be obtained from food
* 20 amino acids
* 2 groups:
1- essential amino acids: we must get them but we can't synthesize them
2- non-essential precursors, can be produced using the essential ones

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Production of amino acids

- Amino acids uses include:

1. **Use in human and animal nutritional supplementation**
   - Some foods such as plant proteins, lacks essential amino acids. ↳ not all amino acids/proteins found in one plant but almost all of them are found in plants in general
   - Animal feeds made from inexpensive plant proteins can be greatly improved with only a small quantity of the limiting amino acids.

2. **Flavor and taste enhancement in foods**
   - Mono-sodium glutamate well-known as a flavoring agent ↳ dangerous, found in soy sauce, ~~indomy~~ اندومي
   - Splenda contain a dipetide formed from aspartic acid and phenylalanine — a sweetner used by diabitics

aspartame (found in diet coke) a sweetner that may have negative effects if it was consumed in a rate more than ~~2,400 mg~~ daily (40 mg/kg)

Production of amino acids

3. **Medical uses:**

مهم

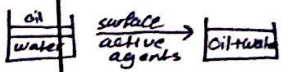
Table 21.2 Therapeutic uses of some amino acids

Amino Acid	*Use*
Ornithine	Used for treating cases of hyperammonemia (excessive ammonia in blood) because it increases urease activity in the liver, thus enhancing urease production.
Arginine	More common than ornithine. 1. Use for ammonia detoxication described above. 2. Arginine is the main component of spermatic proteins and hence administered in cases of male sterility due to low number or weak spermatozoa (*Sterilitas virilis*).
Aspartic acid	1. Used for ammonia detoxification combined with ornithine. 2. Used as carrier for K^+ and Mg^{++} in form of potassium or magnesium aspartate in cases of heart failure, fatigue, etc.
Cysteine and cystine	1. Protect SH-enzymes in the liver from enzyme inhibitors; used for dealing with poisons generally including cyanide poisoning. 2. L-cysteine is used in bronchitis and nasal catarrh.
Gluatamic acid	Important in brain metabolism hence various analogues of glutamic acid are used in treating various neuropathic diseases.
Glycine	Rarely used as a drug, but only as a sweetener in medicines.
Histidine	An amino acid essential for infants but not for adults, it is used in adults for gastric and duodenal ulcers and is administered in cases of anaemia because it helps in haemoglobin regeneration.
Methionine	A sulfur-containing amino acid, it is important in the metabolism of various sulfur-containing compounds in the body. It is also used for detoxification in poisoning by arsenic, chloroform, and benzene derivaties. A derivative of methionine, Vitamin U, is used as an anti-ulcer drug, because it neutralizes histamine which is known to induce ulcer formation.
Tryptophan	Used as an anti-depressant.

Production of amino acids

4. **Use as an industrial synthetic raw materials**

   (a) Surface-active agents : can work as emulsifiers



   (b) Production of polymers from amino acids:
   - Polymers derived from amino acids are used in making synthetic leather, fire-resistant fabrics and anti-static materials.

   (c) Use as cosmetics
   - Amino acids exhibit a buffering action that help maintain normal skin function by regulating pH and a protective action against bacteria.

Production of amino acids

- Amino acids can be produced via four general methods:

(1) **Extraction from natural protein hydrolysates**
(digesting proteins)
(2) **Chemical synthesis**

(3) **Microbiological synthesis:**

A. Semi-fermentation → to digest proteins
B. Use of microbial enzymes or immobilized cells
C. Direct fermentation. : using certain microorganisms to produce certain amino acids

*if proteins weren't digested in the intestines they'll cause allergy

* amino acids:
1- positive
2-negative
3-no charge

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Production of amino acids by the direct fermentation

- The production of amino acids by fermentation was stimulated by the discovery of an efficient L- glutamic acid producer *Corynebacterium glutamincum.*
- The four most widely used bacteria for amino acid production by fermentation are: (produce amino acids in commercial amounts)
    - *Corynebacterium* spp.
    - *Brevibacterium* spp.
    - *Microbacterium* spp.
    - *Arthrobacter* spp.

Production of amino acids by the direct fermentation

batch fermentation (continuous)

fed-batch fermentation

cimultive (total yield) is how much we actually produced

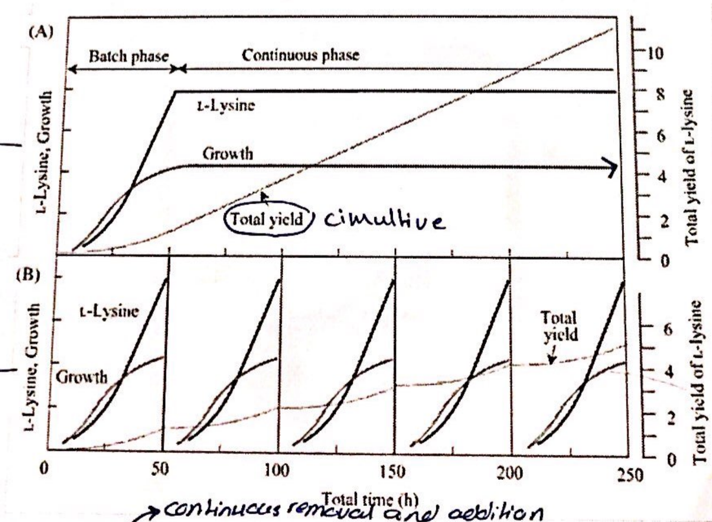


Fig. 3. A Continuous culture for L-lysine production by *C. glutamicum.* B Fed-batch cultures for L-lysine production by *C. glutamicum.* Total yield represents the change of the relative amount of cumulative L-lysine when the final amount of L-lysine after one run of fed-batch culture is defined as 1

Production of Amino Acids by the Direct Fermentation

Production of glutamic acid as an example

→ not minubulated (still how it was naturally produced)

1. Production of amino acids by wild type bacteria
   - Example: Production of glutamic acid by *Corynebacterium glutamicum*
   - Two major means of regulating amino acid synthesis are feedback inhibition and repression.

↓ alone phase

↓ acertain chemical putting pressure on the gene to stop ~~acertain~~ at the whole gene

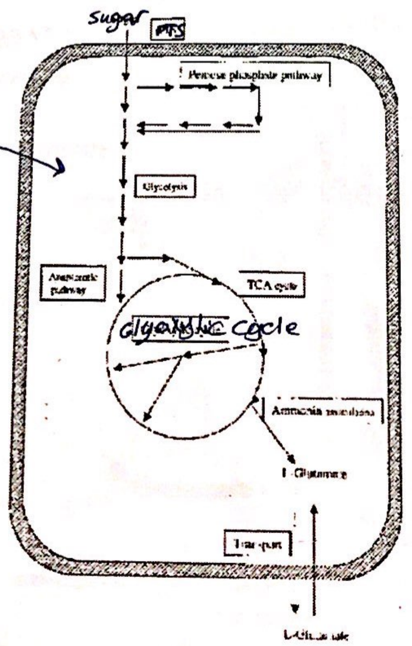
accumilation of product

في سلايد ناقص

Production of Amino Acids by the Direct Fermentation

Production of glutamic acid as an example

if we added a certain nutrient the process will go faster



* can break (elute) the product
change pH
add salt

feedback
inhibition

inhibit the action of

enzyme 1

gene1 gene2 gene3 trp

inhibit the production
of the enzyme

repression inhibition

production of biotin role

manipulation of the metabolic and regulatory process may be made in order to inhance the extracellular yeild of glutamic acid

Glutamic acid is secreted from cells in response to various stress situations and these stresses can be re-created using various treatments such as biotin limitation, addition of penicillin or ~~adding fatty acids~~ explicit fatty acid derivate and ester surfactants. or fermentations employing oleic acid or employing glycerol-auxotroph

Biotin, a water soluble B-complex vitamin, plays a ~~rote~~ vital role in assisting body to metabolize proteins and process glucose.

one of the distinctive features of C. glutamicum is that it needs biotin for growth. It is well accepted that in order to accumulate reasonable amount of extracellular glu, biotin is required in minor quantities

* auxotroph: a bacteria that can't produce certain thing (ex. glycerol)
prototroph: a bacteria that produce all its needs

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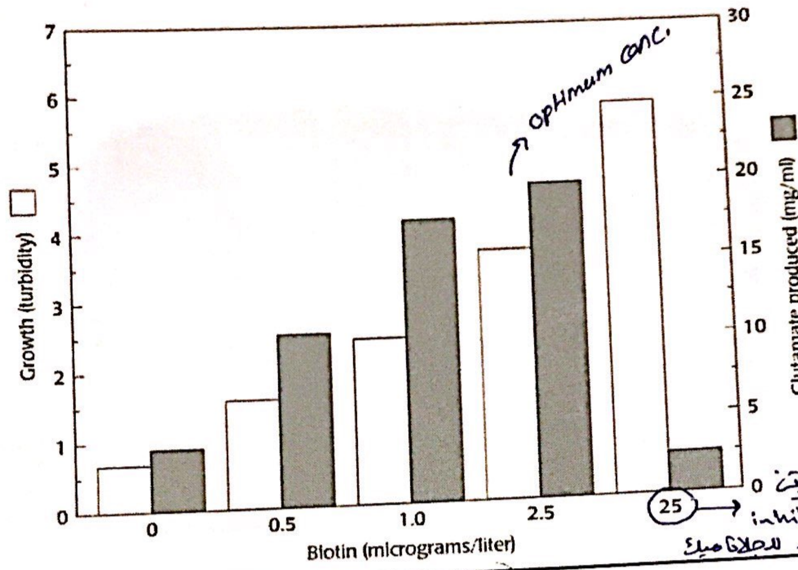
Production of glutamic acid as an example

Biotin role

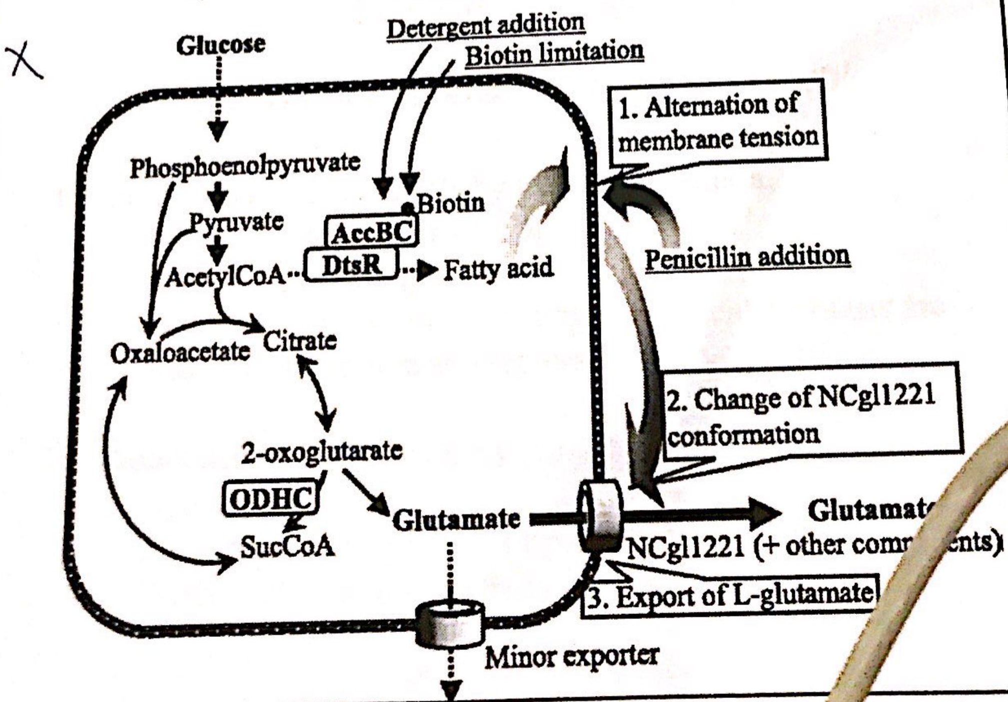


لما زدنا البيوتين كثير صار inhibits لل ~~growth~~ production للجلوتاميك

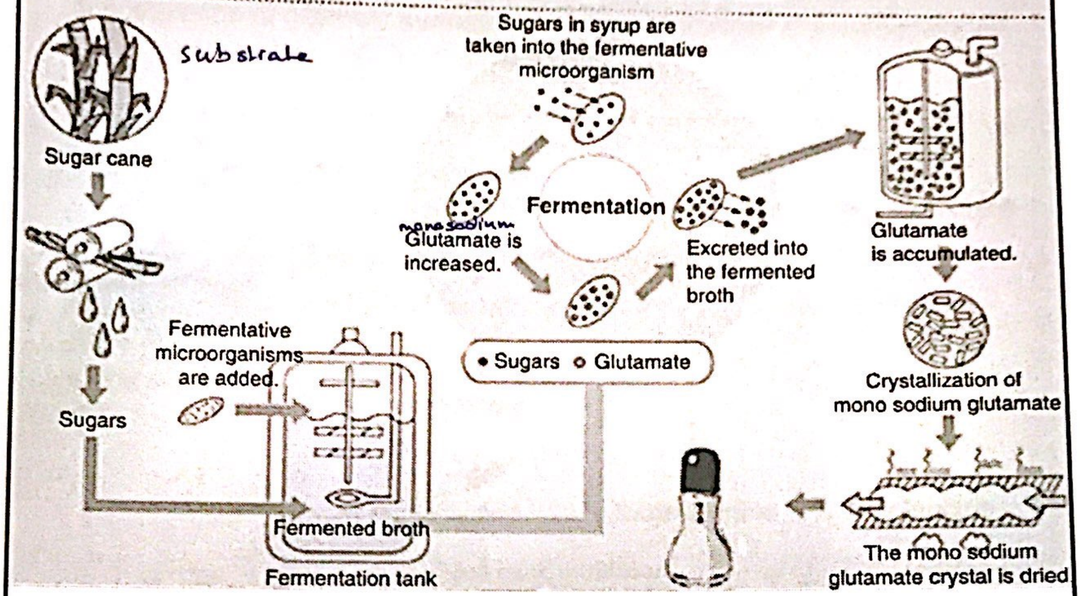
biotin is produced in minor quantities

Production of glutamic acid as an example

Biotin role



Production of mono sodium glutamate by fermentation



Production of Amino Acids by the Direct Fermentation

Production of glutamic acid as an example

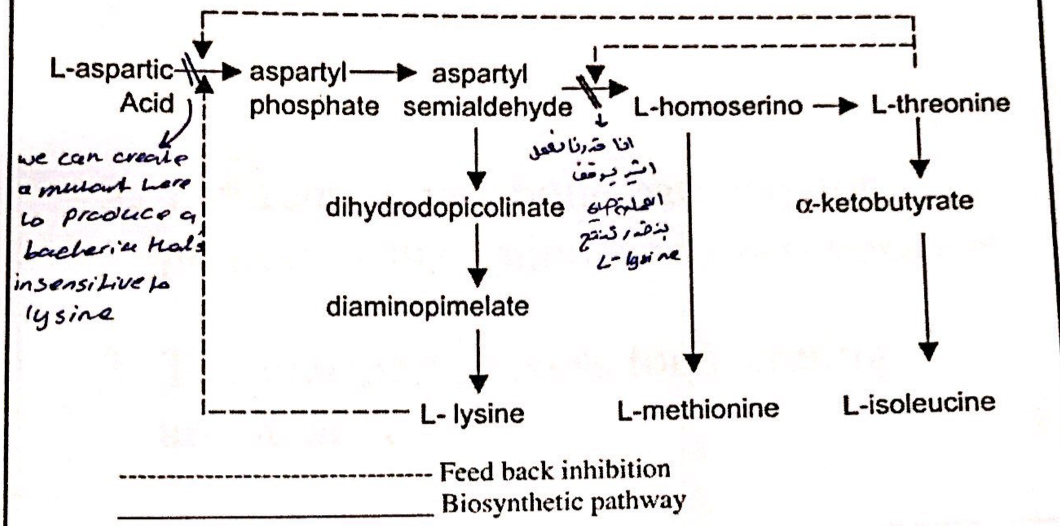
1. Production of amino acids by wild type bacteria
   - Example: Production of glutamic acid by *Corynebacterium glutamicum*
   - Two major means of regulating amino acid synthesis are feedback inhibition and repression.

2. Production of amino acids by mutants

   - Production of amino acids by auxotrophic mutants
   - Production of amino acids by regulatory mutants

Production of amino acids by auxotrophic mutants

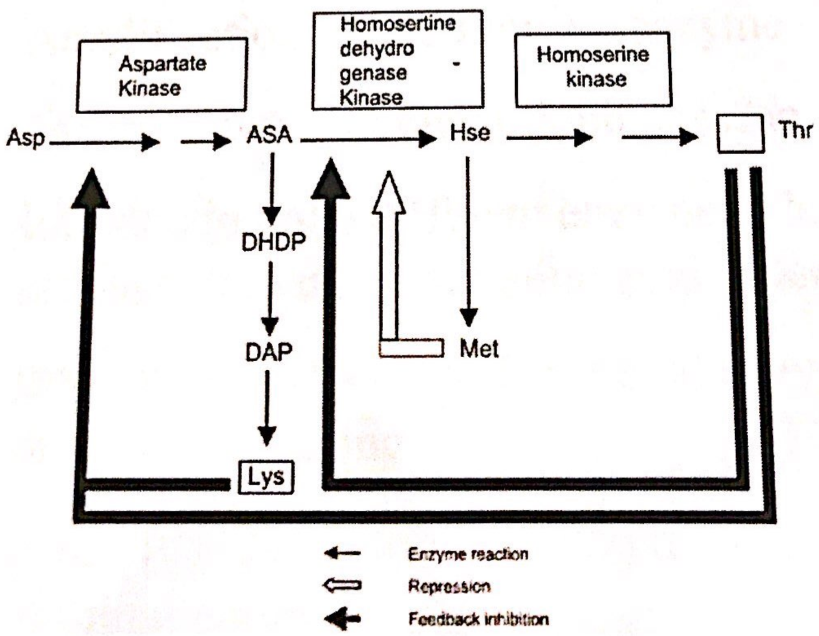
Accumulation of lysine in a mutant auxotrophic strain of Corynebacterium glutamicum



Knowing the pathways can help produce amino acids

Production of amino acids by regulatory mutants

Lysine Biosynthesis in Brevibacteium flavum



ASA = aspartate semino-aldehyde; DADP = dihydrodipicolinate;
Hse = Homoserine; DAP = diaminopimelate

Improvements in the production of amino acids using metabolically engineered organisms

engineering a microorganism to produce a certain enzyme

1. The terminal pathways of the amino acid synthesis
2. The central metabolic pathway for producing the amino acid
3. The transport process for secreting amino acid

Strategies to modify the terminal pathways

1. Amplification of rate limiting enzyme
2. Amplification of branch-point enzyme
3. Introduction of a different enzyme able to produce the same end amino acid (more production)
4. Introduction of a more functional enzyme than the native one
5. Amplification of the first enzyme in the terminal pathway (rate limiting)

$E_1 \rightarrow E_2 \rightarrow E_3 \rightarrow E_4$

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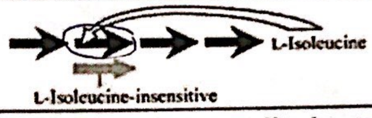
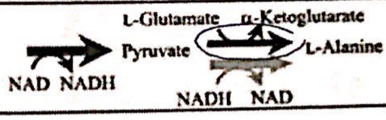
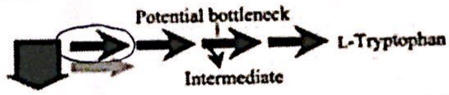
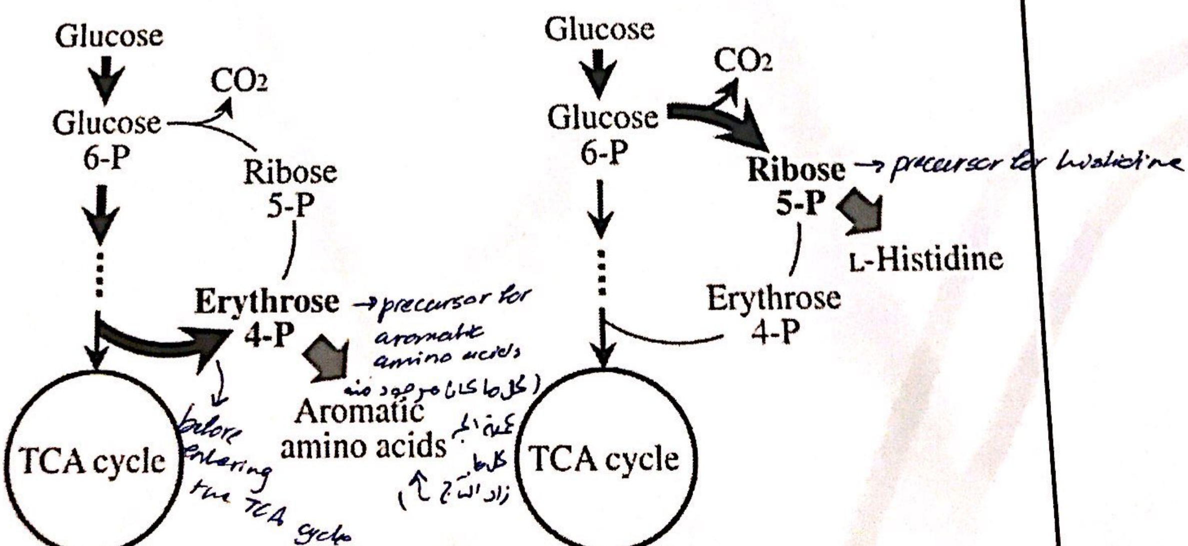
Strategies to modify the terminal pathways

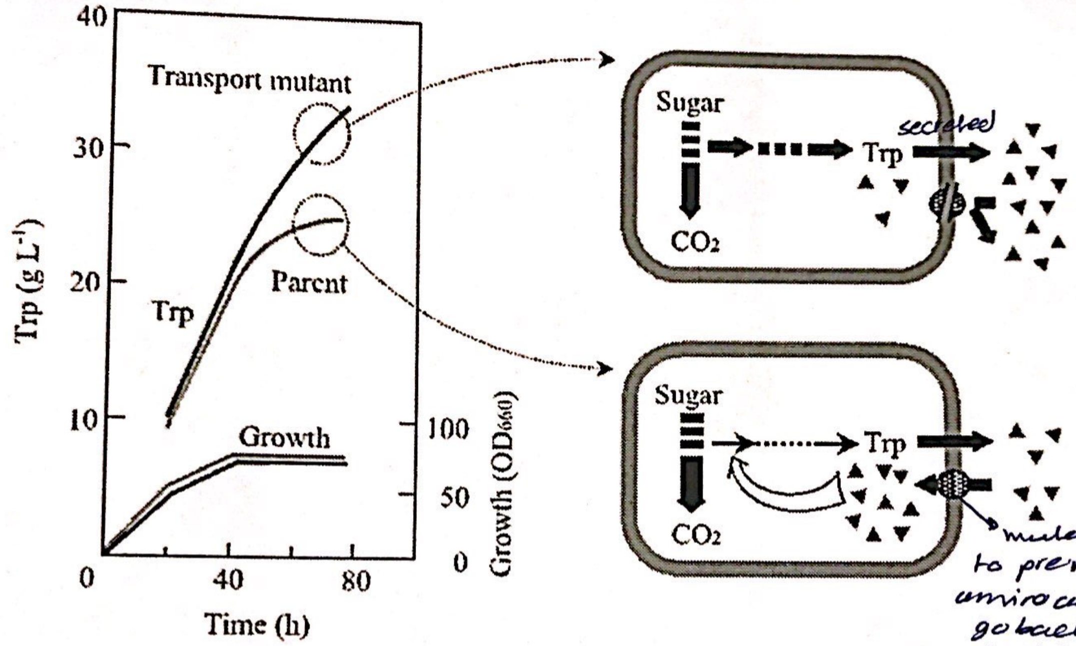
(1) Amplification of rate-limiting enzyme (Removal of bottleneck)	
(2) Amplification of branch-point enzyme (Metabolic conversion)	
(3) Introduction of heterologous enzyme with different control architecture (Bypass of bottleneck reaction)	
(4) Introduction of heterologous enzyme with different catalytic mechanism (Acceleration of key reaction)	
(5) Amplification of the first enzyme in terminal pathways (Augmentation of carbon flow and identification of potential bottleneck)	

Modifying the central metabolic pathway for producing the amino acid



Strategies to increase precursor availability for aromatic and L-Histidine production in *C. glutamicum*

Modifying the transport process for secreting amino acid



Strategies to increase precursor availability for aromatic and L-Histidine production in *C. glutamicum*

mutation to prevent amino acid to go back inside so it'd be easier to collect